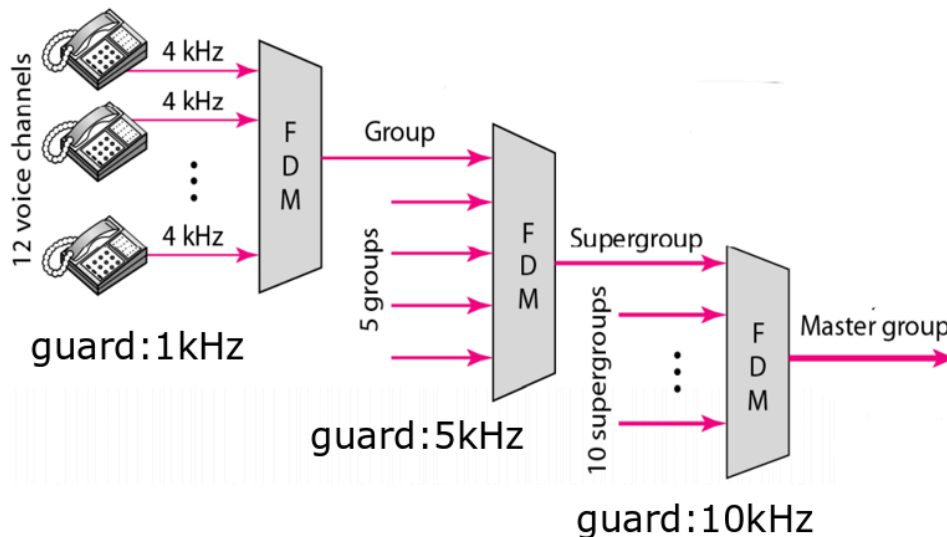


CSE320 Assignment 3
Summer 2025

Q1. Assume we have an output channel bandwidth of 100 KHz starting from 1250 KHz. There are 5 input channels. If the guard band between two channels is 5 KHz, what will be the maximum bandwidth for each channel that we can assign? What will be the maximum frequency?

Q2. The following FDM hierarchy has been used by a telephone company. How many voice channels can be multiplexed together in the master group? What is the required bandwidth for the multiplexing?



Q3. Five channels, two with a bit rate of 240 kbps and three with a bit rate of 180 kbps, are to be multiplexed, with one synchronization bit and two interleaved bits. Answer the following questions.

- Show how this can be achieved.
- What is the input rate and input duration?
- What will be the frame size, frame rate & frame duration?
- What will be the output rate & output duration?
- What will be the output slot duration?

Q4. Five channels are multiplexed using TDM. Each channel can process 1000 pages per 5 seconds. Each page has 20 lines with 40 characters. If 5 lines at a time are to be multiplexed using the concept of TDM with 5 synchronizing bits, then find the following.

- Input rate & input duration
- Frame size, frame rate & frame duration
- Output rate & output duration

Q5. Look at the digital hierarchy in Fig.1 and answer the following questions.

- (a) Calculate the overhead (in terms of bps or kbps) in each of the links of DS-1, DS-2, DS-3, and DS-4.
- (b) If it is known that the entire overhead in the links DS-2 is due to framing bits, and each DS-2 link has a frame rate of 21000 fps, how many framing bits are used in each frame?

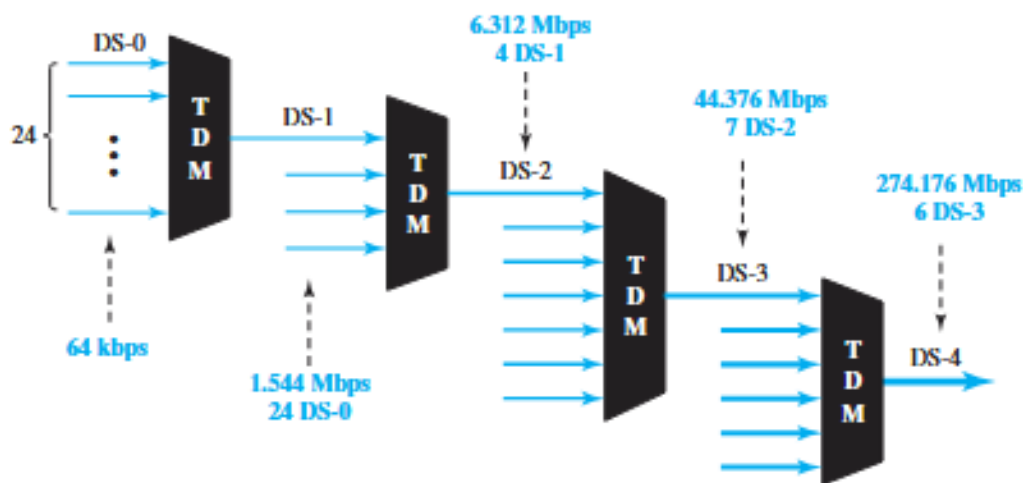


Fig.1: Digital hierarchy for Q5